MACHINE LEARNING AND DEEP LEARNING

Lab 07: Domain Adaptation

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DOMAIN ADAPTATION

- **Domain Adaptation** is a branch of machine learning that focuses on making a model trained on one distribution (the **Source Domain**) perform well on a different, but related, distribution (the **Target Domain**).
- It addresses the **"domain shift" problem** where, for example, a model trained to recognize cars in sunny weather (Source) fails when it encounters cars in heavy snow (Target).
- The goal is to learn features that are **"domain-invariant":** the model should focus on the core characteristics of the data rather than domain-specific noise.

DOMAIN ADAPTATION

- Usually in this framework we have access to **a large number of labeled data** for the **Source Domain** but **few or none labeled data** for the **Target Domain.**
- This issue impose different solutions than standard classification over the Target Domain.
- One of the possible solutions is the **Maximum Mean Discrepancy (MMD)** Loss function: the idea is to force the model to learn features **invariant** to the specific domain, minimizing the differences between the features representations extracted from elements belonging to the Source and elements from the Target.

MMD LOSS – GAUSSIAN KERNEL

- **MMD Loss** relies on the RBF Kernel (or Gaussian Kernel) to **measure the similarities** between the datasets directly from the samples.

$$K(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

- This kernel maps samples into a **high-dimensional space** to measure similarities.
- If x and y are **similar,** the result will be **close to 1.**
- As points get **further apart,** the result decays **toward 0.**

MMD LOSS

- The MMD Loss is then expressed through the following formula:

$$\mathrm{MMD}^2(P,Q) = E[K(X,X')] + E[K(Y,Y')] - 2E[K(X,Y)]$$

- The first term measures the **similarities between X and itself.**
- The second term measures the **similarities between Y and itself.**
- The last term measures the **similarities between X and Y.**
- If X and Y are **similar,** the loss will **collapse to 0.** Otherwise, the loss will be positive.

EXERCISE

- The exercise consists of a Domain Adaptation demo implemented over MNIST and SVHN datasets.
- The goal is to train a model over MNIST dataset and make it able to generalize over SVHN data.
- **Task:** download Domain_Adaption.ipynb and follow the instructions provided in the comments.